16 **B**

$$T = \frac{1}{f} = \frac{1}{5000} = 0.20 \text{ ms}$$

Since T is represented by 2 horizontal squares (2 cm),

$$\text{time base setting} = \frac{0.20}{2} = 0.10 \text{ ms cm}^{-1} = 100 \text{ μs cm}^{-1}$$

17 **D** If the waves meet in phase at O, then path difference = OY – OX = XY = $n\lambda$